

WASP-52b

R-0504

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_230910 · created 2026-07-20T17:54:12+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460197.8779 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.208 +/- 0.061
Transit depth (Rp/Rs)^2	4.34 +/- 2.55 [%]
Orbital Inclination (inc)	81.47 +/- 2.76
Airmass coefficient 1 (a1)	0.54 +/- 0.26
Airmass coefficient 2 (a2)	0.49 +/- 0.36
Scatter in the residuals of the lightcurve fit is	48.15 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.036 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.208 ± 0.061 vs 0.1646 (deviation 0.71σ)
Duration fit vs published	0.036 d vs 0.0758 d (deviation 1.28σ)
Mid-time O–C	-4.0 min
Depth SNR	3.4
Residual scatter	48.15 %

Light curve

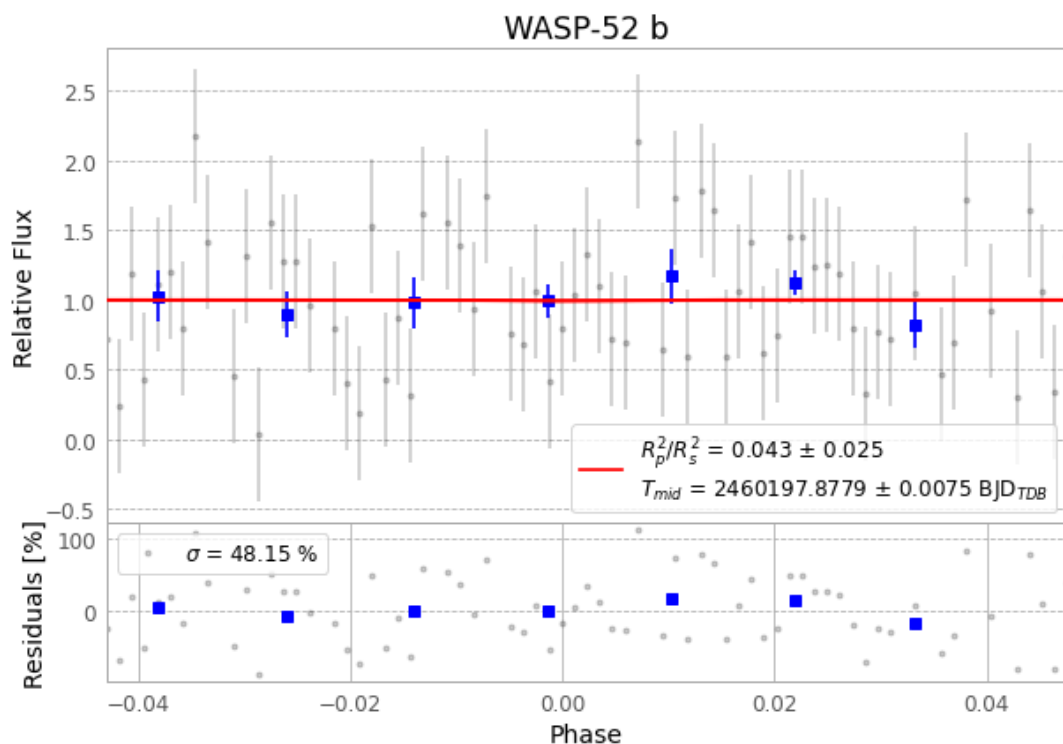


Figure 1 — FinalLightCurve_WASP-52 b_2023-09-10.png (SHA-256 9985e637c16569c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [552, 423], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 28bfa3f1f0e5e606...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

77 FITS frames (51 MB), WASP-52230910070615.FITS → WASP-52230910105415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230910.log (41fc24cfd6ca...), FinalLightCurve_WASP-52 b_2023-09-10.png (9985e637c165...), FinalLightCurve_WASP-52 b_2023-09-10.csv (844717a36acb...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 17:54Z.